

Incognito Intel: The No-Judgment Help Desk

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I. Project Overview

Incognito Intel is a digital platform designed to allow students to ask academic questions anonymously while providing teachers with real-time insights into student struggles. The system removes the fear of embarrassment and promotes a safe learning environment.

The platform includes a key feature called the “*Common Struggles Heatmap*”, which aggregates anonymous questions into data that highlights the most challenging topics for students. This also shows the students that they are not the only one struggling in a certain topic or subject, allowing them to feel more comfortable in asking questions or raising inquiries.

II. Objectives

1. Encourage Participation

Allow students to ask questions freely without fear of judgment.

2. Identify Learning Gaps

Provide teachers with real-time data on commonly misunderstood topics.

3. Improve Teaching Strategies

Help educators adjust lesson plans based on actual student needs.

4. Promote Inclusive Learning

Ensure all students, especially shy or struggling learners, are heard.

III. Key Features

1. Anonymous Question Submission

- Students can submit questions without revealing their identity.
- Removes pressure and fear of being judged.

2. Common Struggles Heatmap

- The system analyzes submitted questions.
- Displays trends such as:
 - “40% of students are asking about Quadratic Equations”
 - “25% are confused about Trigonometric Identities”

3. Teacher Dashboard

- Visual representation (charts/heatmap) of student difficulties.
- Updated in real-time.
- Helps prioritize topics needing review.

4. Topic Categorization

- Questions are grouped automatically (e.g., Algebra, Calculus).
- Makes data easier to interpret.

IV. Innovation

Unlike traditional Q&A systems, *Incognito Intel* focuses on:

- Anonymity + Analytics

Not just answering questions, but detecting patterns in learning difficulties

V. Benefits

For Teachers:

- Real-time feedback on student understanding
- Data-driven lesson planning
- No need to single out struggling students

For Students:

- Safe, judgment-free environment
- Increased confidence in asking questions
- Better understanding of lessons

VI. Implementation Plan

Phase 1: Design

Create UI for student input and teacher dashboard

Phase 2: Development

Build anonymous submission system

Develop data aggregation and heatmap feature

Phase 3: Testing

Pilot testing with a small class

Gather feedback

Phase 4: Deployment

Launch in school or institution

Monitor and improve system

VII. Tools & Technologies

Frontend: HTML, CSS, JavaScript

Backend: Python / Node.js

Database: Firebase / MySQL

Visualization: Charts (heatmaps, graphs)

VIII. Expected Outcome

- Increased student participation
- Improved academic performance
- More responsive and adaptive teaching methods

IX. Conclusion

Incognito Intel transforms the classroom into a data-informed, inclusive environment where students can learn without fear and teachers can teach with clarity. By combining anonymity with analytics, this system bridges the gap between student struggles and effective instruction.

X. Diagram

Class Diagram - UML

Code used:

```
@startuml
title Incognito Intel - Class Diagram
' =====
' Core Classes
' =====

class Student {
    +submitQuestion(content: String, topic: String)
}

class Question {
    -questionID: int
    -content: String
    -topic: String
    -timestamp: DateTime
    +getContent(): String
}

class Teacher {
    +viewDashboard()
    +analyzeHeatmap()
}

class Dashboard {
    +displayHeatmap()
    +displayStatistics()
}

class Heatmap {
    -topicFrequency: Map
    +generateHeatmap()
}

class AnalyticsEngine {
    +aggregateQuestions()
    +calculateFrequency()
}

class Topic {
    -name: String
    -category: String
}

class Database {
```

```

+storeQuestion(q: Question)
+retrieveQuestions()
}
' =====
' Relationships
' =====
Student --> Question : submits
Question --> Topic : categorized as
Teacher --> Dashboard : views
Dashboard --> Heatmap : displays
Heatmap --> AnalyticsEngine : uses
AnalyticsEngine --> Question : analyzes
AnalyticsEngine --> Topic : groups by
Database --> Question : stores
AnalyticsEngine --> Database : retrieves data
@enduml

```

Website used:

<https://plantuml.com/>

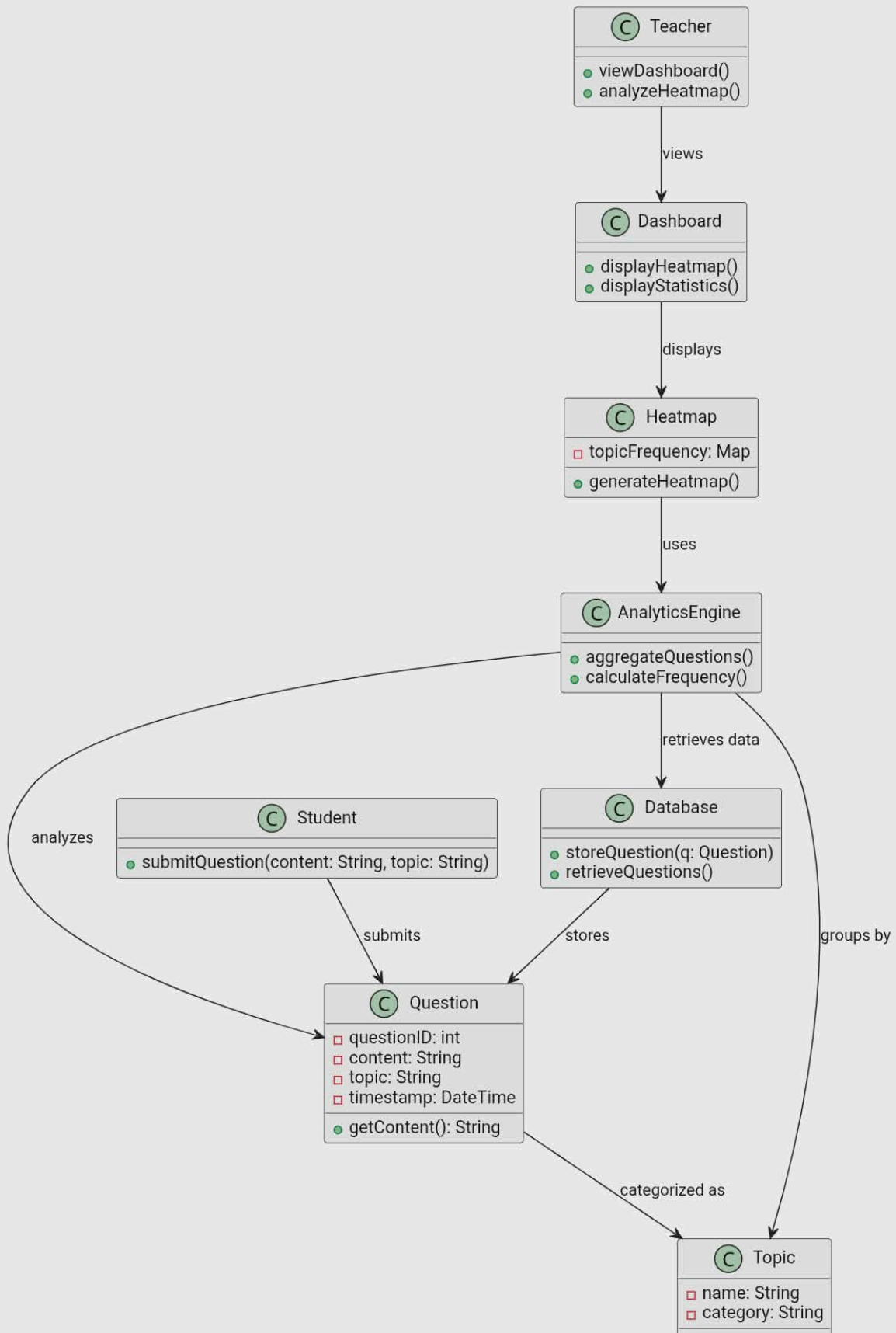
Class Diagram Link/QR Code:



https://www.plantuml.com/plantuml/uml/VLB1Rjim3BthAtXRWoq_8613XsJ3UjXXQtw0aGb5W2ov8jt1BVhlao9Ry8oiESMFxv6FUhndmIHZxvKuyGGFmKGRdCJyJyZ37WuUcU7eq2RibVe0ds_-Sk4G4rKWyVzvobmRFihOKH1uLG0VUJptJdwDnE9Yk3CnIAxfJ4eks4yWSN1c-TofjwN9ehXssJ_DNmz73Iv8WJQT2jHwDS3rMOVzeE68GgVyMJnPaaDLt-qOlKq-4PeBfMh_sT6V8_Bb735rTxi2OK0_IT0FGkbn05XJDcBLTeu7ZrCZhh17GN4ipl1QFpFXjItpFL5UFPX9mqySglT02UNM-A_5MEdvBLWN9f0QnFPB5WUbAiHWzwCFkDjnBhLgGolFWBsj7vJapKsfGbXg_K5pyZpO9QOgEN-f5kYr_69ifYU_t7rzk9_YOFL-R53U_Tu79z-_sNDX8qr3jarQ1Ihsjgc7ToBzG1ibek85DMcMeeny2g8QNUCjCmHyjgWKfvcuc6aOdL5jwOhITsczTCsnJ7WU4ygVRmsvrB1BURD8c69Gg6BeDAtLFenjx_1G00

Illustration:

Incognito Intel - Class Diagram



Flowchart

Code used:

```
@startuml
title Incognito Intel - System Flowchart
start
:Student opens platform;
:Submit anonymous question;
:Enter topic/category;
if (Valid input?) then (Yes)
:Store question in Database;
:Send data to Analytics Engine;
:Aggregate questions;
:Update topic frequency;
:Generate Heatmap;
:Update Teacher Dashboard;
:Teacher views dashboard;
:Identify common struggles;
:Adjust lesson plan;
else (No)
:Display error message;
endif
stop
@enduml
```

Website used:

<https://plantuml.com/>

Flowchart Link/QR Code:



https://www.plantuml.com/plantuml/uml/HP6nRlCm38HtFmN7PFZnxzdG14ZQPkcIja17HQ9b5HBfYbG3lttf66qcc_VTqJlv4tL_LMycT9iq89_8SAldRcsA6Vt2UHB70O-Qh7vpRTQAtn-QiBI0fy8WaC6QdFTUodKcxbAJWY6agt0l-6eecfcstEP9Y1UKn-V_U_AKQkasLI3wjtrE0H6EJXpNeW0lh3vHr1s0Vuuf_VSm41wVkuWGjF7Ea0C4aK8OzkInfyW97YedCOevzZ1MZqtiNkUbleyL_meGZwYaR9JqlN9oliCt6fyMDfjuJh-ZyWdK6aU72heQ5_ihV2Oy2uGwDdUO5fNu2pwKmWMXjCMQK9ReFdqqKh1O3jb4_o7RCWh5xujeP34fCdm5gvGZ6ZY_PxI27rysLux7PMs45_0000OIKq-4PeBfMh_sT6V8_Bb735rTxI2OK0_IT0FGkbn05XJDeB_LTeu7ZrCZhh17GN4ipl1OFpFXjltPFL5UEFPX9mqySglT02UNM-A_5MEdvBLWNg9f0OnFPB5WUbAiHWzwCFkDjnBhLgGo_lFWBsj7vJapKsfgbXg_K5pyZpO9OOgEN-f5kYr_69ifYU_t7rzk9_YOFL-R53U_Tu79z-_sNDX8qr3jarO1lHsjgc7ToBzGlibek_85DMcMeeny2g8ONUCjCmHyjgWKfvcuc6aOdL5jwOhITsczTCsnJ7WU4ygVRmsvrB1BURD8c69Gg6BeDAtLFenx_1G00

Illustration:

